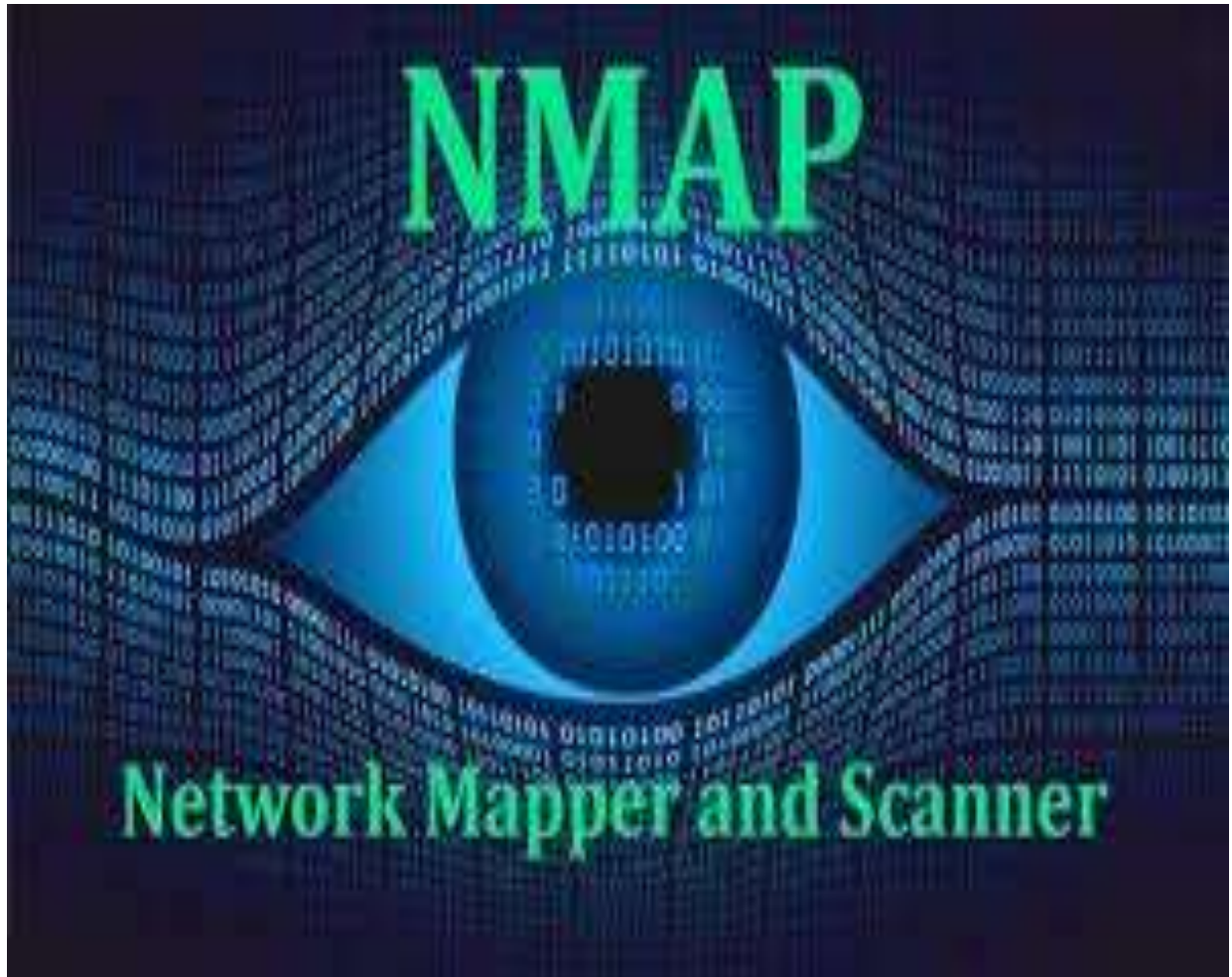


Network Mapper Discover Hosts



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Introduction & Overview

Nmap has become one of the most popular tools in network scanning by leaving other scanners behind. Many times, network administrators secure hosts in some organizations using firewalls or intrusion prevention systems. However, these security measures can sometimes block scanning

attempts due to the preset rules used to filter network traffic. In Nmap, a pentester can easily make use of alternate host discovery techniques to prevent this from happening. It consists of certain

features that make the network traffic a little less suspicious. Hence, let us look at various techniques of Host Discovery.

Ping Sweep

Let's begin with scanning the entire network by using the Ping sweep scan (-sP).

```
nmap -sP 192.168.1.0/24
```

```
kali@kali:~# nmap -sP 192.168.1.0/24
Starting Nmap 7.91 ( https://nmap.org ) at 2020-11-20 05:43 EST
Nmap scan report for dsldevice.lan (192.168.1.1)
Host is up (0.0012s latency).
MAC Address: 18:45:93:69:A5:10 (Taicang T&W Electronics)
Nmap scan report for 192.168.1.3
Host is up (0.00030s latency).
MAC Address: 8C:EC:4B:71:C5:DE (Dell)
Nmap scan report for 192.168.1.4
Host is up (0.024s latency).
MAC Address: 2A:84:98:9F:E5:5E (Unknown)
Nmap scan report for 192.168.1.5
Host is up (0.012s latency).
MAC Address: 30:24:32:1F:89:AC (Intel Corporate)
Nmap scan report for 192.168.1.8
Host is up (0.0058s latency).
MAC Address: 44:CB:8B:C2:20:DA (LG Innotek)
Nmap scan report for 192.168.1.12
Host is up (0.00027s latency).
MAC Address: 00:0C:29:78:20:90 (VMware)
Nmap scan report for 192.168.1.108
Host is up (0.00017s latency).
MAC Address: 00:0C:29:C8:9C:50 (VMware)
Nmap scan report for 192.168.1.9
Host is up.
Nmap done: 256 IP addresses (8 hosts up) scanned in 23.67 seconds
kali@kali:~#
```

When you closely observe the packets in Wireshark, you will notice that network scans send only ARP packets.

arp					
No.	Time	Source	Destination	Protocol	Length Info
64	0.555456987	TaicangT_69:a5...	Dell_71:c5:de	ARP	60 who has 192.168.1.3? Tell 192.168.1.1
65	0.555463118	Dell_71:c5:de	TaicangT_69:a5:d0	ARP	60 192.168.1.3 is at 8c:ec:4b:71:c5:de
200	1.589157098	TaicangT_69:a5...	VMware_b2:bb:77	ARP	60 who has 192.168.1.9? Tell 192.168.1.1
210	1.589161561	VMware_b2:bb:77	TaicangT_69:a5:d0	ARP	42 192.168.1.9 is at 00:0c:29:b2:bb:77
228	1.974212283	VMware_b2:bb:77	Broadcast	ARP	42 Who has 192.168.1.17? Tell 192.168.1.9
220	1.974288490	VMware_b2:bb:77	Broadcast	ARP	42 Who has 192.168.1.27? Tell 192.168.1.9
230	1.974336247	VMware_b2:bb:77	Broadcast	ARP	42 Who has 192.168.1.3? Tell 192.168.1.9
231	1.974396943	VMware_b2:bb:77	Broadcast	ARP	42 Who has 192.168.1.47? Tell 192.168.1.9
232	1.974433312	VMware_b2:bb:77	Broadcast	ARP	42 Who has 192.168.1.5? Tell 192.168.1.9
233	1.974456164	Dell_71:c5:de	VMware_b2:bb:77	ARP	60 192.168.1.3 is at 8c:ec:4b:71:c5:de
234	1.974466332	VMware_b2:bb:77	Broadcast	ARP	42 Who has 192.168.1.6? Tell 192.168.1.9
235	1.974494541	VMware_b2:bb:77	Broadcast	ARP	42 Who has 192.168.1.7? Tell 192.168.1.9
236	1.974541930	VMware_b2:bb:77	Broadcast	ARP	42 Who has 192.168.1.8? Tell 192.168.1.9
237	1.974675204	VMware_b2:bb:77	Broadcast	ARP	42 Who has 192.168.1.10? Tell 192.168.1.9
238	1.974694993	VMware_b2:bb:77	Broadcast	ARP	42 Who has 192.168.1.11? Tell 192.168.1.9

▶ Frame 64: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0
 ▶ Ethernet II, Src: TaicangT_69:a5:10 (18:45:33:69:a5:10), Dst: Dell_71:c5:de (8c:ec:4b:71:c5:de)
 ▶ Address Resolution Protocol (request)

```

0000  8c ec 4b 71 c5 de 15 45 93 99 a5 10 08 08 00 01  --Kq--E-i-----
0010  08 00 08 04 00 01 18 45 93 69 a5 10 c8 a8 01 01  -----E-i-----
0020  00 00 00 00 00 00 c8 a8 01 03 00 00 00 00 00 00  -----
0030  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  -----
  
```

Note: Working of **-sP** and **-sn** is the same.

Let us try the same by using **the no port scanning (-sn)** option. In this option, we are also

using -packet-trace option which will enable you to see the detailed packet transfer without making use of Wireshark. Here you can observe the ARP packets being received.

```
nmap -sn 192.168.1.0/24 --packet-trace
```



```
kali@kali:~# nmap -sn 192.168.1.0/24 --packet-trace
Starting Nmap 7.91 ( https://nmap.org ) at 2020-10-20 05:48 EST
SENT (0.0687s) ARP who-has 192.168.1.1 tell 192.168.1.9
SENT (0.0688s) ARP who-has 192.168.1.2 tell 192.168.1.9
SENT (0.0689s) ARP who-has 192.168.1.3 tell 192.168.1.9
SENT (0.0690s) ARP who-has 192.168.1.4 tell 192.168.1.9
SENT (0.0691s) ARP who-has 192.168.1.5 tell 192.168.1.9
SENT (0.0692s) ARP who-has 192.168.1.6 tell 192.168.1.9
SENT (0.0692s) ARP who-has 192.168.1.7 tell 192.168.1.9
SENT (0.0693s) ARP who-has 192.168.1.8 tell 192.168.1.9
SENT (0.0694s) ARP who-has 192.168.1.10 tell 192.168.1.9
SENT (0.0695s) ARP who-has 192.168.1.11 tell 192.168.1.9
RCVD (0.0690s) ARP reply 192.168.1.3 is-at 8C:EC:48:71:C5:DE
RCVD (0.0699s) ARP reply 192.168.1.3 is-at 18:45:93:69:A5:10
SENT (0.0730s) ARP who-has 192.168.1.14 tell 192.168.1.9
SENT (0.0731s) ARP who-has 192.168.1.15 tell 192.168.1.9
SENT (0.0731s) ARP who-has 192.168.1.16 tell 192.168.1.9
SENT (0.0732s) ARP who-has 192.168.1.17 tell 192.168.1.9
RCVD (0.0791s) ARP reply 192.168.1.4 is-at 2A:8A:98:9F:E5:5E
RCVD (0.0796s) ARP reply 192.168.1.5 is-at 30:34:32:1F:89:AC
SENT (0.0820s) ARP who-has 192.168.1.20 tell 192.168.1.9
SENT (0.0822s) ARP who-has 192.168.1.21 tell 192.168.1.9
SENT (0.0823s) ARP who-has 192.168.1.22 tell 192.168.1.9
SENT (0.0824s) ARP who-has 192.168.1.23 tell 192.168.1.9
SENT (0.1699s) ARP who-has 192.168.1.26 tell 192.168.1.9
SENT (0.1703s) ARP who-has 192.168.1.27 tell 192.168.1.9
SENT (0.1705s) ARP who-has 192.168.1.28 tell 192.168.1.9
SENT (0.1708s) ARP who-has 192.168.1.29 tell 192.168.1.9
SENT (0.1710s) ARP who-has 192.168.1.30 tell 192.168.1.9
SENT (0.1712s) ARP who-has 192.168.1.31 tell 192.168.1.9
```

Now that we have identified ARP packets in the network, we can use the `--disable-arp-ping` option. In this case, you will observe that four packets are sent instead.

Disable-arp-ping

To disable the ARP discovery, Nmap provides this option.

```
nmap -sn 192.168.1.108 --disable-arp-ping
```

```
root@kali:~# nmap -sn 192.168.1.108 --disable-arp-ping
Starting Nmap 7.91 ( https://nmap.org ) at 2020-11-20 05:58 EST
Nmap scan report for 192.168.1.108
Host is up (0.00027s latency).
MAC Address: 00:0C:29:C8:9C:50 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds
root@kali:~#
```

And you will see that the ARP packets are not visible

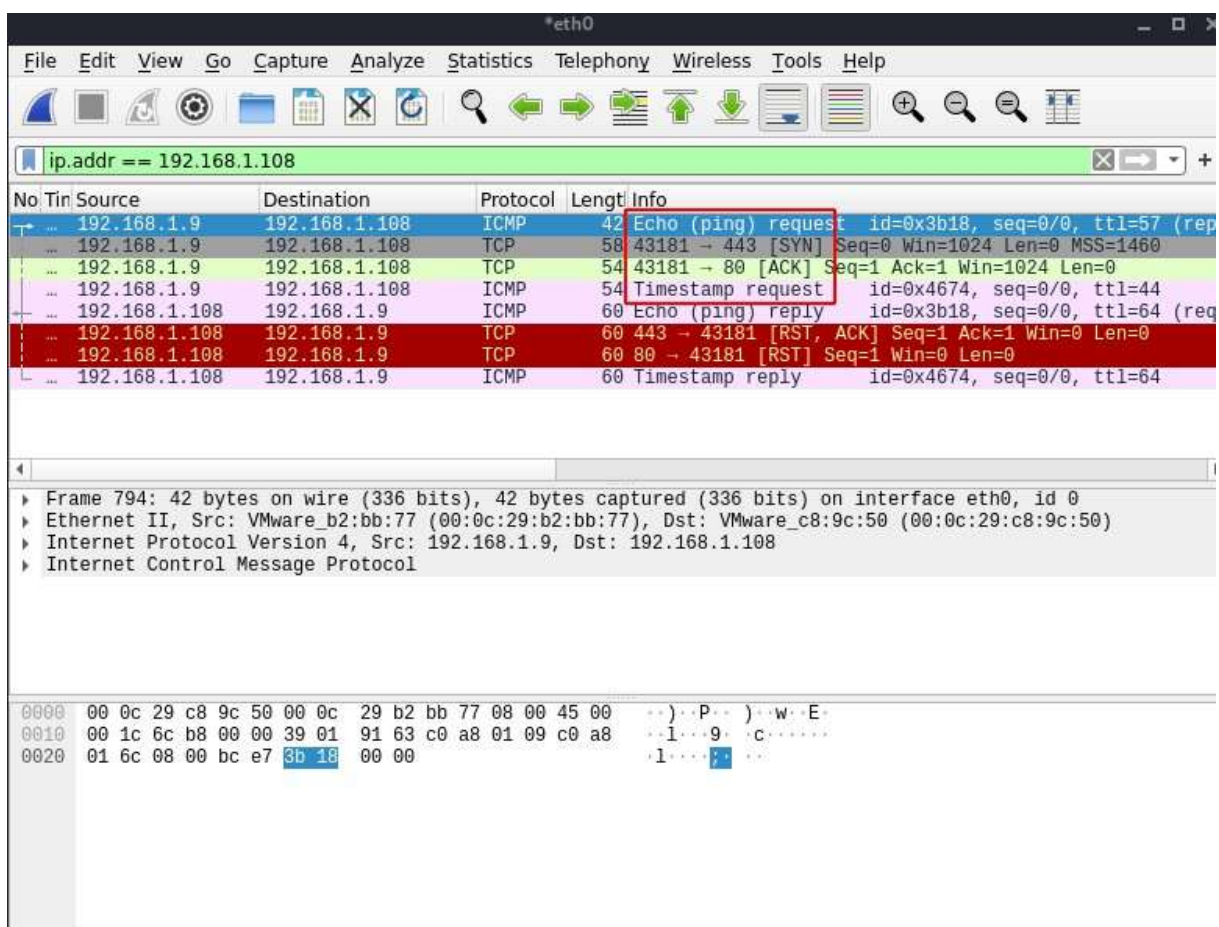
Note: When you scan the Local Network with Nmap, Nmap sends an ARP packet with every scan. If you scan an external network, Nmap sends the following request packets when `--disable-arp-ping` is used:

ICMP echo request (Type 8)

ICMP timestamp request(Type 13)

TCP SYN to port 443

TCP ACK to port 80



You can also make use of **--send-ip** option to get the same results as in the step above.

send-ip

```
nmap -sn 192.168.1.108 --packet-trace --send-ip
```

```

root@kali:~# nmap -sn 192.168.1.108 --packet-trace --send-ip
Starting Nmap 7.91 ( https://nmap.org ) at 2020-11-20 05:55 EST
SENT (0.0588s) ICMP [192.168.1.9 > 192.168.1.108 Echo request (type=8/code=0)]
SENT (0.0589s) TCP 192.168.1.9:43573 > 192.168.1.108:443 S ttl=58 id=30850 iplen=40
SENT (0.0589s) TCP 192.168.1.9:43573 > 192.168.1.108:80 A ttl=55 id=52947 iplen=40
SENT (0.0589s) ICMP [192.168.1.9 > 192.168.1.108 Timestamp request (type=13/code=0)]
RCVD (0.0590s) ICMP [192.168.1.108 > 192.168.1.9 Echo reply (type=0/code=0) id=30850]
NSOCK INFO [0.1030s] nsock_ioc_new2(): nsock_ioc_new (IOD #1)
NSOCK INFO [0.1030s] nsock_connect_udp(): UDP connection requested to 192.168.1.108
NSOCK INFO [0.1030s] nsock_read(): Read request from IOD #1 [192.168.1.1:53] (type READ)
NSOCK INFO [0.1030s] nsock_write(): Write request for 44 bytes to IOD #1 EID 27
NSOCK INFO [0.1030s] nsock_trace_handler_callback(): Callback: CONNECT SUCCESS
NSOCK INFO [0.1030s] nsock_trace_handler_callback(): Callback: WRITE SUCCESS for
NSOCK INFO [0.1090s] nsock_trace_handler_callback(): Callback: READ SUCCESS for
NSOCK INFO [0.1090s] nsock_read(): Read request from IOD #1 [192.168.1.1:53] (type READ)
NSOCK INFO [0.1090s] nsock_ioc_delete(): nsock_ioc_delete (IOD #1)
NSOCK INFO [0.1090s] nevent_delete(): nevent_delete on event #34 (type READ)
Nmap scan report for 192.168.1.108
Host is up (0.00024s latency).
MAC Address: 00:0C:29:C8:9C:50 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds

```

Host Discovery is the most fundamental step in Information Gathering. It provides accurate results regarding active ports and IP addresses in a network.

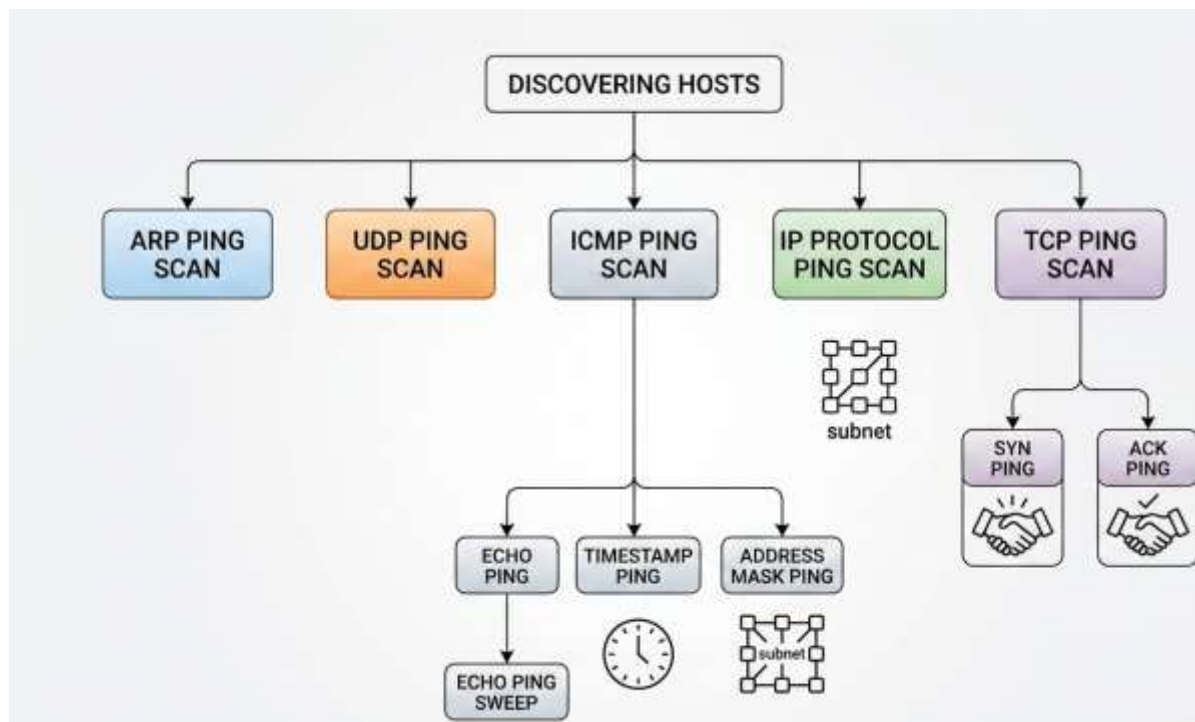
TCP Flags

First, let's get to know the basics about the communication Flags in TCP. The TCP header mainly consists of six flags that manage the connection between the systems and provide instructions to them. Each flag is of 1 bit and hence the size of TCP Flags is 6 bits. Now let us briefly understand each flag.

FLAG	DESCRIPTION
SYN	It stands for Synchronize. It assists in notifying when a new sequence number is transmitted. The SYN flag usually represents the Three-Way Handshake.
ACK	It stands for Acknowledgement. It notifies the status of transmission of packets and also assists in identifying the what sequence number to expect next.
RST	It stands for Reset. This flag shows when there is any error in that connection and sets the flag to 1 and the connection is broken.
URG	It stands for Urgent. This flag usually commands to process the packets as soon as possible.
FIN	It stands for Finish. This flag is set as 1 to indicate no further transmission of packets.
PSH	It stands for Push. It is used to start and end data transfer and prevent occurrence of buffer deadlocks.

Types of Scans

To discover the hosts in the network, various ping scan methods can be used.



TCP SYN Ping Scan

It is a method of host discovery which helps in looking for discovering if the ports are open and to also make sure if it matches the rules of the firewall. The Pentester can hence, send an empty SYN flag to the target to check where it is alive. Multiple ports can be defined in this scan type.

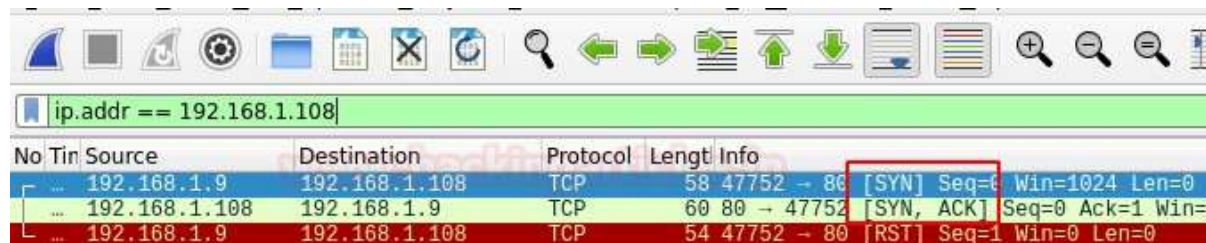


The `-sP` command in Nmap is specifically designed for discovering online hosts. On the other hand, the SYN Ping (`-PS`) command sends a TCP SYN packet to specific ports. If the port is closed, the host responds with an RST packet. If the requested ports are open, the host replies with a TCP SYN/ACK, followed by a reset packet to terminate the connection.

```
nmap -sn -PS 192.168.1.108 --disable-arp-ping
```

```
root@kali:~# nmap -sn -PS 192.168.1.108 --disable-arp-ping
Starting Nmap 7.91 ( https://nmap.org ) at 2020-11-20 07:13 EST
Nmap scan report for 192.168.1.108
Host is up (0.00030s latency).
MAC Address: 00:0C:29:C8:9C:50 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds
```

The packets captured using Wireshark can be overserved

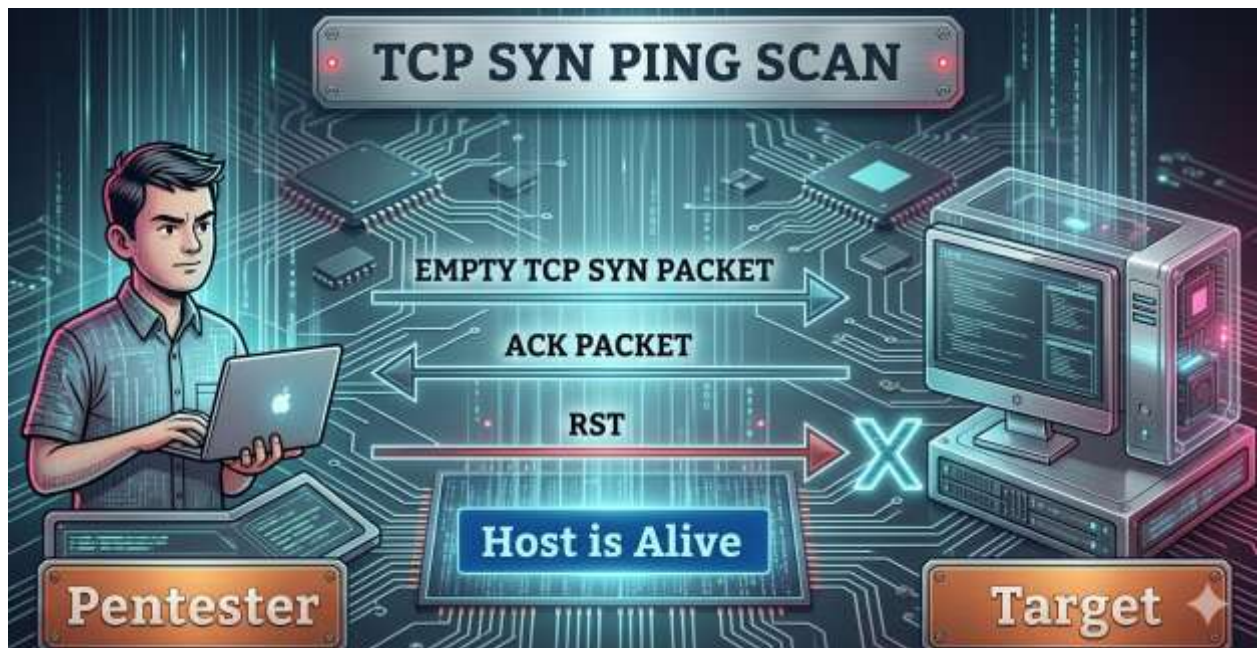


No	Tin	Source	Destination	Protocol	Length	Info
...	...	192.168.1.9	192.168.1.108	TCP	58	47752 → 80 [SYN] Seq=0 Win=1024 Len=0
...	...	192.168.1.108	192.168.1.9	TCP	60	80 → 47752 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
...	...	192.168.1.9	192.168.1.108	TCP	54	47752 → 80 [RST] Seq=1 Win=0 Len=0

The advantage of TCP SYN Ping scan is that the pentester can get the active/inactive status of the host without even creating a connection and hence it does not even create a log in the system or the network.

TCP ACK Ping Scan

It is a method of host discovery that is similar to TCP SYN Ping scan, but it slightly differs. This scan also makes use of Port 80. The pentester sends an empty TCP packet to the target, and as there is no connection between them, it will receive an Acknowledgement packet and will then reset and terminate the request

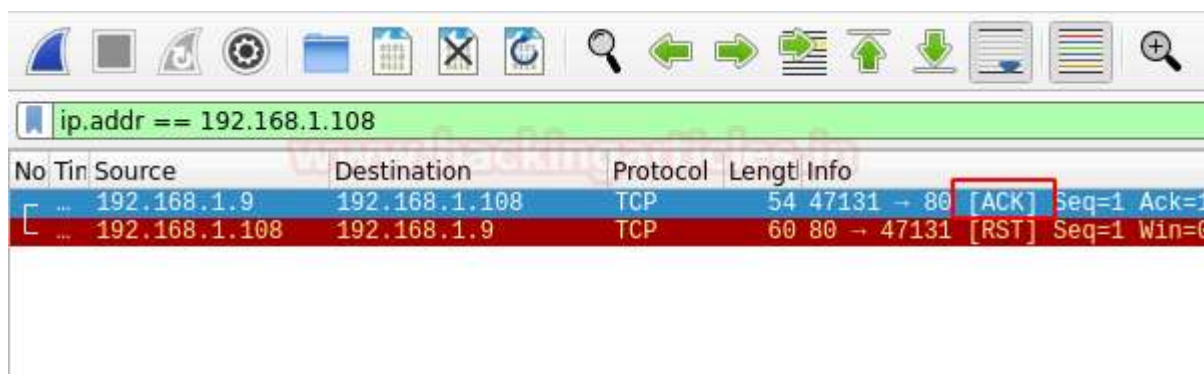


Additionally, this command helps to determine how the target responds. It also checks if firewalls block the SYN packets or ICMP echo requests, as is common with many modern firewalls.

```
nmap -sn -PA 192.168.1.108 --disable-arp-ping
```

```
root@kali:~# nmap -sn -PE 192.168.1.108 --disable-arp-ping
Starting Nmap 7.91 ( https://nmap.org ) at 2020-11-20 07:15 EST
Nmap scan report for 192.168.1.108
Host is up (0.00039s latency).
MAC Address: 00:0C:29:C8:9C:50 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds
root@kali:~#
```

You can observe the captured packets in Wireshark here.



This scan might work to get past the firewall in this situation because some systems set up firewalls to prevent SYN ping packets.

ICMP Echo Ping Scan

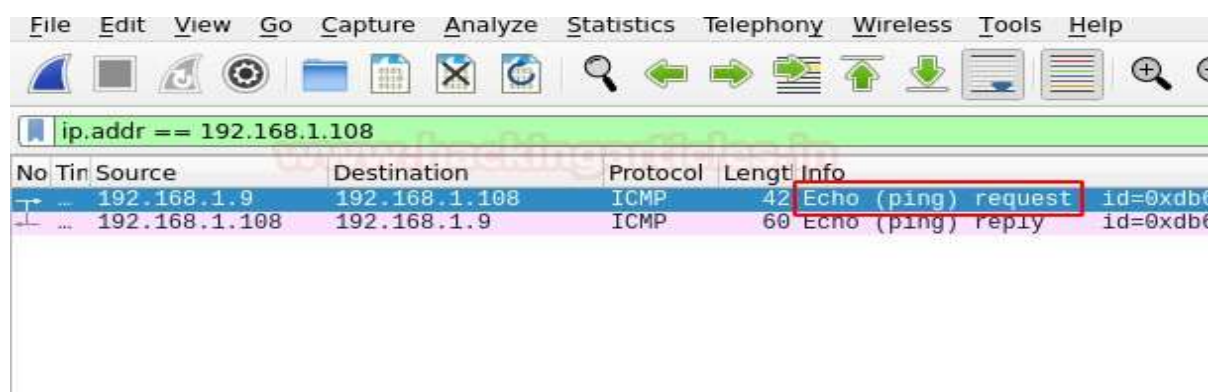
The ICMP Ping scan gathers information about target systems, making it distinct from port scanning. The pentester sends an ICMP ECHO request to the target and receives an ICMP ECHO reply in response.



```
nmap -sn -PE 192.168.1.108 --disable-arp-ping
```

```
root@kali:~# nmap -sPE 192.168.1.108 --disable-arp-ping
Starting Nmap 7.91 ( https://nmap.org ) at 2020-11-20 07:15 EST
Nmap scan report for 192.168.1.108
Host is up (0.00039s latency).
MAC Address: 00:0C:29:C8:9C:50 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds
root@kali:~#
```

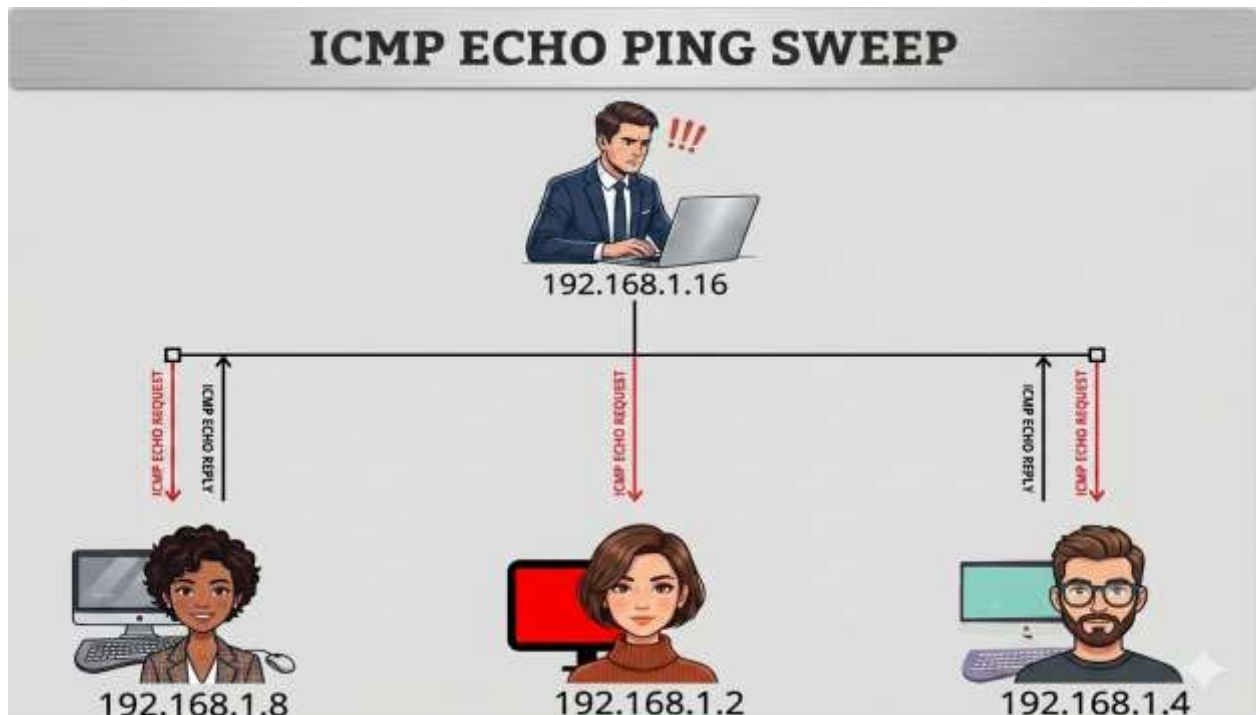
The packets captured in Wireshark can be observed.



ICMP ECHO Ping Sweep

It resembles Echo Ping Scan and scans the active hosts from a given range of IP addresses. It sends ICMP requests to a huge number of targets, and if a particular target is alive, then it returns an ICMP reply.

```
nmap -sn -PE 192.168.1-10
```

ICMP Address Mask Scan

It is an older method of ICMP ECHO ping scanning. It gives out the information about the system and its subnet mask.

```
nmap -sn -PM 192.168.1.108 --disable-arp-ping
```

```
root@kali:~# nmap -sPE 192.168.1.108 --disable-arp-ping
Starting Nmap 7.91 ( https://nmap.org ) at 2020-11-20 07:15 EST
Nmap scan report for 192.168.1.108
Host is up (0.00039s latency).
MAC Address: 00:0C:29:C8:9C:50 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds
root@kali:~#
```

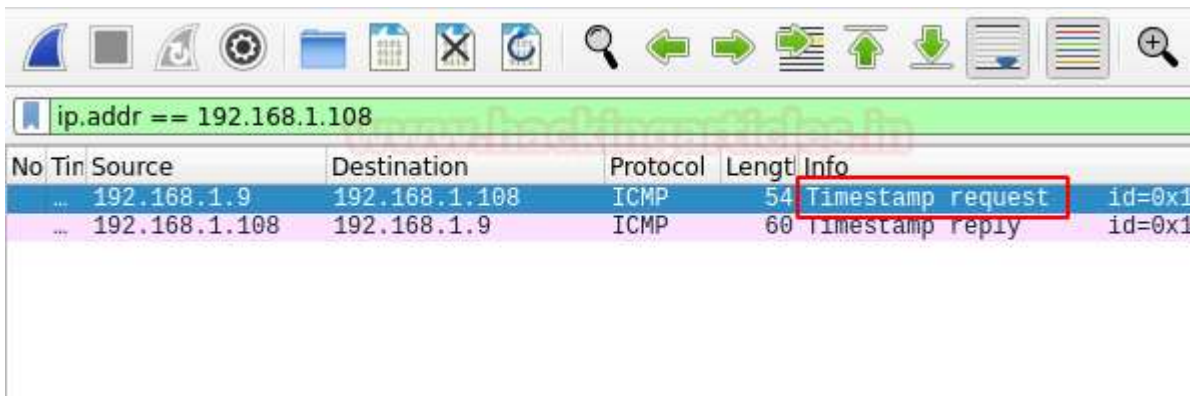
ICMP ECHO Timestamp scan

The pentester can adopt this technique in a particular condition when the system admin blocks the regular ICMP timestamp. It is usually used in synchronization of time.

```
nmap -sn -PP 192.168.1.108 --disable-arp-ping
```

```
root@kali:~# nmap -sn -PP 192.168.1.108 --disable-arp-ping
Starting Nmap 7.91 ( https://nmap.org ) at 2020-11-20 07:17 EST
Nmap scan report for 192.168.1.108
Host is up (0.00059s latency).
MAC Address: 00:0C:29:C8:9C:50 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
root@kali:~#
```

The packets captured using Wireshark can be observed.



The image shows a Wireshark packet capture interface. The filter bar at the top is set to `ip.addr == 192.168.1.108`. The packet list shows two ICMP packets. The first packet is a 'Timestamp request' from 192.168.1.9 to 192.168.1.108. The second packet is a 'Timestamp reply' from 192.168.1.108 to 192.168.1.9. The 'Info' column for the first packet is highlighted with a red box.

No	Tin	Source	Destination	Protocol	Length	Info
...	...	192.168.1.9	192.168.1.108	ICMP	54	Timestamp request id=0x1
...	...	192.168.1.108	192.168.1.9	ICMP	60	Timestamp reply id=0x1

UDP Ping Scan

The UDP Ping Scans uses a highly uncommon default port number 40125 to send packets to the target. It is the same to TCP Ping scan. The Pentester will send the UDP Packets to the target and if there is a response in return, which means that the host is alive, or else it is offline

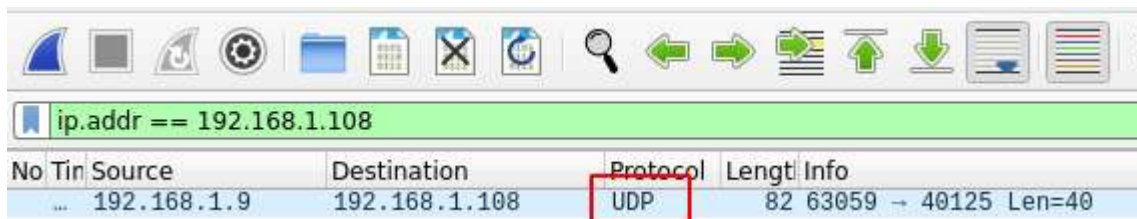


The advantage of UDP scan is that it can detect systems that have firewalls with strict TCP rules while leaving UDP rules at ease.

```
nmap -sn -PU 192.168.1.108 --disable-arp-ping
```

```
root@kali:~# nmap -sn -PU 192.168.1.108 --disable-arp-ping
Starting Nmap 7.91 ( https://nmap.org ) at 2020-11-20 12:06
Nmap scan report for 192.168.1.108
Host is up (0.00032s latency).
MAC Address: 00:0C:29:C8:9C:50 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds
```

You can observe the packets sent using Wireshark.



No	Time	Source	Destination	Protocol	Length	Info
...	...	192.168.1.9	192.168.1.108	UDP	82	63059 → 40125 Len=40

IP protocol ping scan

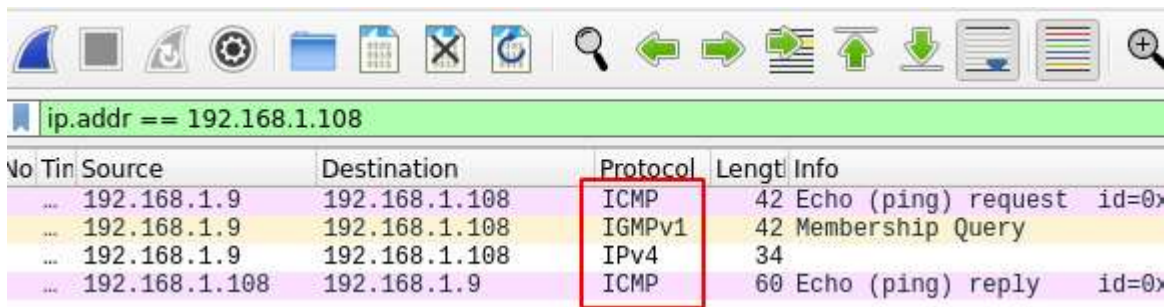
In this method, the pentester sends various packets using different IP protocols and hopes to get a response in return if the target is alive.



```
nmap -sn -PO 192.168.1.108 --disable-arp-ping
```

```
root@kali:~# nmap -sn -PO 192.168.1.108 --disable-arp-ping
Starting Nmap 7.91 ( https://nmap.org ) at 2020-11-20 12:07 EST
Nmap scan report for 192.168.1.108
Host is up (0.00040s latency).
MAC Address: 00:0C:29:C8:9C:50 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
root@kali:~# nmap
```

The packets captured can be observed using Wireshark.



No	Time	Source	Destination	Protocol	Length	Info
...	...	192.168.1.9	192.168.1.108	ICMP	42	Echo (ping) request id=0x...
...	...	192.168.1.9	192.168.1.108	IGMPv1	42	Membership Query
...	...	192.168.1.9	192.168.1.108	IPv4	34	
...	...	192.168.1.108	192.168.1.9	ICMP	60	Echo (ping) reply id=0x...

No ping scan

In this method, host discovery is completely skipped. The pentester can use it to determine active machines for heavier scanning and to increase the speed of the network.

```
nmap -sn -PN 192.168.1.108 --disable-arp-ping
```

```
root@kali:~# nmap -sn -PN 192.168.1.108 --disable-arp-ping
Host discovery disabled (-Pn). All addresses will be marked 'up' and scan
Starting Nmap 7.91 ( https://nmap.org ) at 2020-11-20 12:10 EST
Nmap scan report for 192.168.1.108
Host is up up.
Nmap done: 1 IP address (1 host up) scanned in 0.01 seconds
root@kali:~#
```

ARP ping scan

In this method, the system sends ARP packets to all the devices in the network, even though these devices remain invisible due to the firewall. ARP ping scans are generally considered more efficient than other host discovery methods. Moreover, they

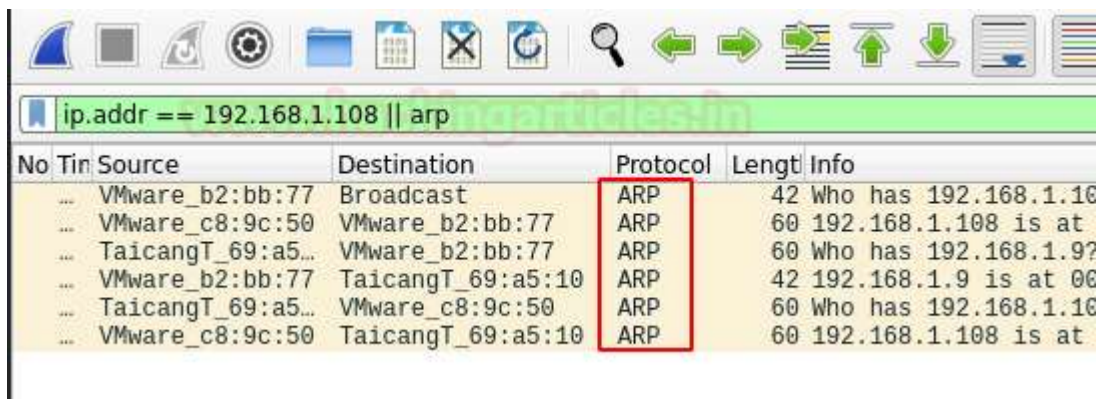


primarily focus on system discovery and also provide latency details.


```
nmap -sn -PR 192.168.1.108
```

```
root@kali:~# nmap -sn -PR 192.168.1.108
Starting Nmap 7.91 ( https://nmap.org ) at 2020-11-20 12:12 EST
Nmap scan report for 192.168.1.108
Host is up (0.00029s latency).
MAC Address: 00:0C:29:C8:9C:50 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.27 seconds
root@kali:~#
```

You can see the packets being captured in Wireshark.



No	Time	Source	Destination	Protocol	Length	Info
...		VMware_b2:bb:77	Broadcast	ARP	42	Who has 192.168.1.108
...		VMware_c8:9c:50	VMware_b2:bb:77	ARP	60	192.168.1.108 is at
...		TaicangT_69:a5:...	VMware_b2:bb:77	ARP	60	Who has 192.168.1.9?
...		VMware_b2:bb:77	TaicangT_69:a5:10	ARP	42	192.168.1.9 is at 00
...		TaicangT_69:a5:...	VMware_c8:9c:50	ARP	60	Who has 192.168.1.108
...		VMware_c8:9c:50	TaicangT_69:a5:10	ARP	60	192.168.1.108 is at

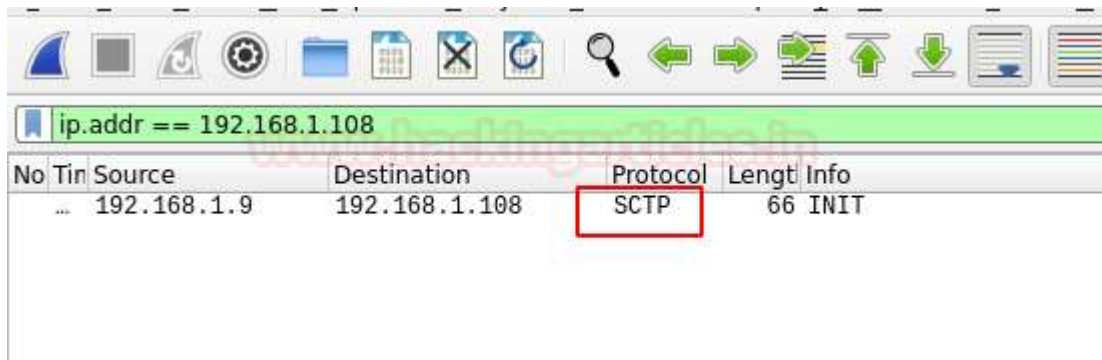
SCTP INIT Ping

It sends SCTP packet containing a minimal INIT chunk. Its default destination port is 80. The INIT chunk provides a suggestion to the remote system that the pentester is attempting to establish an association.

```
nmap -sn -PY 192.168.1.108 --disable-arp-ping
```

```
root@kali:~# nmap -sn -PY 192.168.1.108 --disable-arp-ping
Starting Nmap 7.91 ( https://nmap.org ) at 2020-11-20 12:13 EST
Nmap scan report for 192.168.1.108
Host is up (0.00030s latency).
MAC Address: 00:0C:29:C8:9C:50 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
root@kali:~#
```

The packets that are captured can be observed.



No	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.9	192.168.1.108	SCTP	66	INIT

Traceroute

Users typically perform traceroutes after finishing a scan. At this stage, you can leverage the information from the scan results to determine the port and protocol that will reach the target.

```
nmap -sn --traceroute 8.8.8.8
```

```
root@kali:~# nmap -sn --traceroute 8.8.8.8
Starting Nmap 7.91 ( https://nmap.org ) at 2020-11-20 11:38 EST
Nmap scan report for dns.google (8.8.8.8)
Host is up (0.0014s latency).

TRACEROUTE (using port 80/tcp)
HOP RTT      ADDRESS
1   1.85 ms dsldevice.lan (192.168.1.1)
2   1.57 ms dns.google (8.8.8.8)
```

Network Mapper